

Band of excellence — Army porch & basketball θ (memo)

Purpose: Print-friendly anchor for why we drop own $\hat{A} = 0$ on congestion plots, how that relates to basketball θ , and which porch panels to trust for the story.

Date: 2026-09-24

Status: Design agreed in chat; code prep for Vantage upload (not yet run on Army).

One-sentence claim

Congestion binds in the band where scarce distinction is actually at stake — top blocks and promotion among officers senior raters are really sorting — not in the full captain cross-section from $\hat{A} = 0$ upward.

Three filters (do not confuse)

Filter	What it does	Where it lives
Missing SNR history	<code>tb_ratio_fwd_snr</code> is NaN (no SNR-checked evals)	Already excluded from Panel 2 finite histograms
Own $\hat{A} = 0$ (proposed)	Drop ratees with last-snapshot SNR TB ratio exactly 0 (had SNR evals, never top-blocked)	Plot scripts 7–8–9 only — no feather rebuild
Peer $\hat{A} = 0$ (Run 3)	Drop peers with <code>tb_ratio = 0</code> from pool mean/size	Cell 5 / <code>POOL_EXCLUDE_PEER_TB_ZERO</code>

SNR ratio rule (Cell 4): numerator = SNR top blocks; denominator = SNR-checked evals only. No SNR box $\rightarrow$ does not count toward $\hat{A}$ (NaN, not zero).

Filter order for panels 7, 8, 9

When the own- $\hat{A} = 0$ toggle is on:

1. **Require any finite OER/pool check column** (rename from mislabeled “zero-OER” filter).
2. `pool_size_snr_fwd  $\geq$  3` at last captain snapshot.
3. **Exclude own** `tb_ratio_fwd_snr == 0.0` (finite zero only).
4. Finite plot variables / dropna as today.

Pool of six, two ever-TB’d: All six can pass min-pool if each row has `pool_size  $\geq$  3`. Only the two with $\hat{A} > 0$ enter the congestion cohort. The pond still “made the cut”; we are not requiring ≥ 3 ever-TB’d peers in the plot sample.

Basketball $\theta \rightarrow$ Army band

Basketball (MBB)	Army
θ : viability threshold — “in the draft conversation” (from drafted perf / K/N)	$\hat{A} > 0$: at least one SNR top block in career cumulative through last snapshot
Panel 7 CCT: fixed $\hat{A}$ band (e.g. $z \in [1, 2]$), vary pool LOO	Same Act II probe on Army feather
Panel 8 elite pond: top ~20% $\hat{A}$, upper LOO tail	“Superstar pool” / thick elite pond
Congestion read: upper LOO bins , not full HERO left tail	Same: panels 7–8 upper bins , not Panel 9 lower T bins

Interpretive line for Alex: Senior raters at rating time are disproportionately comparing officers who might receive **scarce top blocks**, not averaging over officers with zero TB history. Ever-TB’d is a coarse empirical gate for “in the game”; CCT/elite gates are the **band of excellence** proper.

Which porch panel for which question

Panel	Question	Band of excellence?
2	Distributions of $\hat{A}$, T (last snapshot)	Full finite sample (before/after filter optional later)
5	H_sort / pool overlap (assortativity)	Optional same own- $\hat{A} = 0$ filter when testing
7	CCT: fixed high $\hat{A}$, promotion vs pool LOO bins	Yes — primary congestion test
8	Elite pond: top $\hat{A} \times$ upper LOO	Yes — thickest elite ponds
9	HERO: unconditional promotion vs pool x	Descriptive full porch ; lower bins may drift if own- $\hat{A} = 0$ dropped — do not over-read

Lead with 7–8 for mechanism. Panel 9 stays transparency; conditional story lives where θ logic lives.

Dropping own $\hat{A} = 0$ — tradeoffs

Pros

- Focuses on officers **competing on the top-block margin**.
- Aligns Army deck with basketball conditional / θ design.
- Removes a large mass at $\hat{A} = 0$ that is a different estimand (“SNR-rated, never distinguished”).

Caveats

- **Lower $\hat{T}$ / LOO bins on Panel 9** can look odd or “inflated” after exclusion — composition shift, not necessarily new physics.
- A six-officer pool may contribute only one–two officers to 7–8–9 after the filter — report **n excluded** in footers/JSON.
- This is **not** re-computing pools; feather pool metrics unchanged.

Methods one-liner (copy-ready)

Officers required `pool_size_snr_fwd ≥ 3` at last captain snapshot. Congestion panels further restrict to own SNR TB ratio > 0 (≥ 1 senior-rater top block in cumulative career through last snapshot). Pool size uses observed pond structure; excluded officers are omitted from promotion-vs-pool plots only. Primary inference: conditional bands (CCT, elite pond), mirroring basketball viability θ and conditional CCT.

Workflow reminder (Army)

- Code edits on **Mac (Dropbox)**; **no Army data runs locally**.
- **Old PDE**: runs possible; **uploads blocked** — transcribe by hand if urgent.
- **Vantage**: starts from admin frozen snapshot; **upload Mac files** to catch up, then run and **screenshot** plots/errors.

Planned code (when Vantage is live)

1. Rename HERO “zero-OER” → `require_any_oer_metric` (~35–45 lines, 2 files).
2. Add `EXCLUDE_OWN_TB_ZERO_LAST` (default **off**) on panels 7–8–9 (~25–35 lines).
3. Later: before/after BDP for $\hat{A}$, $\hat{T}$, LOO (Panels 2–3); checklist glossary row.

Thread log

Date	Note
2026-09-24	Memo from Agent chat: band of excellence, θ parallel, filter order, panel roles